

Jialin Huang

Backend Engineer

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Fullstack engineer, eight years across six industries, lately on the identity layer of a multi-tenant cybersecurity SaaS: SAML federation into Cognito, the OIDC token layer (JWKS validation, custom claims), claims-based RBAC enforced with Casbin, and the auth-event audit pipeline, in Go and Node. On my own I go deeper: a Go URL shortener that splits OLTP from OLAP, and a RAG pipeline with its own eval harness. I build APIs the client actually wants to consume.

EDUCATION **National Chengchi University (NCCU)** · B.A. in Economics

EXPERIENCE

MammothCyber Senior Software Engineer

Apr 2025 - Jul 2026 · Cybersecurity · Taipei (Hybrid)

Full-stack across the web portal of a multi-tenant enterprise security platform, a Chromium-based secure browser, and Go services; identity & access domain (SAML, OIDC/JWT, RBAC, audit, policy config)

NESTJS APOLLO GRAPHQL GOLANG GIN POSTGRESQL DOCKER KUBERNETES JENKINS AWS LAMBDA AWS COGNITO
PLAYWRIGHT

Identity, Access & Audit

- Worked across the auth stack (Go + Node): added tenant-level forced re-authentication (SAML), and threaded an OS / device-posture field across the login-policy service, the Cognito audit pipeline, and the browser and extension clients
- Claims-based RBAC end to end: permissions computed in the Cognito PreTokenGeneration Lambda and compressed into the id_token, enforced by a Casbin guard behind a GraphQL @rbac directive
- Auth-event audit pipeline in Go: Cognito CloudTrail events polled from SQS, classified (password / MFA / device failures), enriched from PostgreSQL with an LRU cache, shipped to Logstash
- Added Google / Apple social login (OAuth) to Cognito alongside the existing SAML, reworking the Lambda triggers that map each social user to its tenant

Policy Configuration Console

- Built the policy-configuration backend in NestJS: rule and content-profile APIs, RBAC-gated, with audit logging and an N+1 fix that collapsed a multi-call save into a single request

Secure Browser & Extension (Chromium)

- Shipped a login / session-visibility feature across every client of a Chromium-based enterprise browser, extending each platform's native bridge in C++ (Android / Desktop) and Objective-C++ (iOS), plus a Go fetch backend and a React UI

Platform & Build Infrastructure

- Drove a monorepo dependency cleanup on npm: root overrides and targeted upgrades (Express v5, Apollo Server v5, ESLint v9) cut vulnerabilities 74 → 13 (-82%), clearing every high & critical, and stray local packages 22 → 0
- **npm to pnpm**: Migrated the monorepo to pnpm workspaces with a staged rollout (shamefully-hoist bridge, declare every phantom dep, flip to strict isolation), cutting installs 3-10x and making per-worktree installs cheap so parallel AI agents can each build in their own git worktree
- Cut two Docker images ~80% in three steps: per-workspace pnpm deploy --prod instead of copying the root node_modules, misdeclared prod deps fixed, then both servers bundled (Remix SSR.noExternal + esbuild, NestJS nest build --webpack) so the runtime image ships no node_modules. Remix web app 1.88 GB → 264 MB, NestJS API 1.95 GB → 490 MB

Bunastar Frontend Engineer

Nov 2024 - Apr 2025 · Gaming · Taipei (Hybrid)

Real-time multiplayer card game for the Southeast Asian market; custom Canvas engine and WebSocket sync

WEBSOCKET

Southeast Asian Gaming Platform - Pok Deng

- Real-time card sync over WebSocket

Confetti Full Stack Engineer
Dec 2022 - Oct 2024 · Gaming · Taipei (Hybrid)
N/A FOR THIS VIEW

SCMP Frontend Engineer
Oct 2021 - Nov 2022 · Media · Hong Kong (Remote)
South China Morning Post: high-traffic international news platform, delivered fully remote
NODE.JS **GRAPHQL**

scmp.com ↗

- Maintained the CMS-driven content rendering pipeline (Node.js, GraphQL)

Futuresai Frontend Engineer
Nov 2020 - Oct 2021 · Finance · Taipei (Hybrid)
SIGNALR **FIREBASE AUTH** **FIREBASE**

Trading Dashboard + Admin + Auth

- Monorepo rewrite with SignalR for real-time trading-data streaming
- Firebase auth middleware with custom membership logic, plus SMS and LINE push notifications

Dentalltw Frontend Engineer
Jul 2018 - Nov 2020 · Healthcare · Taipei (Hybrid)
Built a 0-to-1 clinic product: a desktop app for Taiwan and a React web app for the Vietnam market
C#/.NET **PYTHON/FLASK** **DOCKER** **GCP**

CI/CD Pipeline

- Set up CI/CD to build and push Docker images to GCR, using a Go service and GitLab scripts

PROJECTS

[URL Shortener](#) ↗

One redirect touches five systems: PostgreSQL for link mappings, Redis for live counters, Kafka for the click pipeline, ClickHouse for analytics, and a Go/Gin service tying them together.

GO **GIN** **POSTGRESQL** **REDIS** **KAFKA** **CLICKHOUSE**

- Clicks flow through Kafka in 500-event / 2s batches across 4 writer goroutines, so analytics writes never block a redirect
- Write-behind counter: redirects INCR in Redis, a 5s worker drains to PostgreSQL; reads merge the durable total with the live delta, streamed to a dashboard over Server-Sent Events

221b

RAG over my own writing: blog posts, AI-agent sessions, and config files in one corpus, queried in Chinese, answered in my English voice.

PYTHON **CHROMADB** **SENTENCE-TRANSFORMERS** **MCP**

- Chinese question → multilingual-e5 retrieval over ~4,000 chunks → bge cross-encoder rerank → answer grounded in my English writing.
- Eval harness reports Recall@5 and MRR against ground-truth articles, so a model or prompt swap shows up as a number.

WRITING

- [N+1 isn't a database problem. It's a shape.](#) ↗
- [HTTP 1.1, 2, and 3: what broke and what fixed it](#) ↗
- [Five ways to reach a service in Kubernetes](#) ↗
- [A packet's walk through a VPC, in AWS](#) ↗
- [Stopping XSS and CSRF: who's responsible for what](#) ↗
- [Hiroki's Ark](#) ↗